Do LLMs Learn From Usage After Training?

Conversation excerpt, Thursday, October 15, 2026

Prompt

do any of the LLM AI models, including those by other vendors, adjust the model settings based on usage after the learning phase

Reply

No. None of the major production LLMs (Claude, GPT, Gemini, Llama, Mistral, Grok and others) change their weights in response to your usage. Once a model ships, its parameters are frozen. Every user hits the same fixed weights, and nothing you type alters them in real time.

Usage still influences behavior in several other ways, and that is probably what you have noticed.

1. Usage feeds the next model, not the current one

Vendors may use conversation data, ratings (thumbs up/down), and flagged failures as training data for future versions. Each vendor has its own policy, usually with opt-in or opt-out controls for consumer accounts. API and enterprise traffic is typically excluded by default. The improvement arrives as a new release weeks or months later, not as live learning.

2. Silent version updates

A name like "gpt-x" or "claude-sonnet" can point to a newer snapshot over time. From the outside this can look like the model learning, but it is a deployment swap. Pinning a dated model ID avoids it.

3. Per-user adaptation without weight changes

This is the closest thing to "adjusting to you":

- **Memory and custom instructions:** stored facts and preferences that get added to the prompt.
- **Retrieval:** pulling in your documents or past chats at query time.
- **In-context learning:** the model adapts within a single conversation, then forgets when the context is gone.

These change the input, not the model. Everyone else using the same model is unaffected.

4. Customer fine-tuning

OpenAI, Google (Vertex AI), AWS Bedrock, Azure, and the open-weight ecosystem let you train a private copy or adapter (such as LoRA) on your own data. That does change weights, but it is a deliberate offline job you run. The model does not do it automatically from usage.

5. Settings like temperature

Sampling parameters are set per request by the application. Nothing learns or tunes them from usage.

Why vendors don't do live learning

- **Safety:** users could deliberately poison the model.
- **Privacy:** one person's data could leak into another's answers.

- **Catastrophic forgetting:** updates can degrade existing skills.
- **Reproducibility:** a fixed model behaves predictably and can be evaluated before release.

True continual learning, where a deployed model keeps updating its weights, is an active research area but not in mainstream products as of now. The continual learning survey listed below covers the techniques, and the HackerNoon piece discusses why production systems use memory as a workaround instead.

For teaching purposes, a useful framing is that today's LLMs are a compiled binary plus a growing set of runtime configuration (prompts, memory, retrieval). Recompiling happens only at the vendor, on their schedule.

Sources

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